Vidush Singhal

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Education

- **Purdue University**, West Lafayette, IN, USA
  Ph.D. in Electrical and Computer Engineering — May 2021 – December 2026 (Expected)
- **Purdue University**, West Lafayette, IN, USA
  M.S. in Electrical and Computer Engineering — May 2021 – May 2023
- **Purdue University**, West Lafayette, IN, USA
  B.Sc. in Computer Engineering — Aug 2017 – May 2021

Research Interests

I am a Ph.D. student in Electrical and Computer Engineering at Purdue University working on compiler techniques for improving the performance of programs over irregular recursive datatypes. My research interests include compilers, programming languages, program analysis, static analysis, data layout optimization, locality optimization, SIMD vectorization, automatic parallelization, and compilers for GPUs. I have experience building compiler infrastructure in Gibbon, LLVM, Clang, and MLIR/Torch-MLIR.

Experience

- **Microsoft Research** [] — *May 2025 – August 2025*
  Research Intern — Redmond, WA, USA
  - Developed **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
  - Implemented a **Clang-to-Clang** source transformer (**~7K LOC of C++**) that emits Pulse code for F* verification.
  - Built a C benchmark suite and validated generated verification conditions on representative annotated C programs.
  - Implemented and verified components of **DPE** (**Dice Protection Environment**) written in C through C2Pulse.
- **Lawrence Livermore National Laboratory** [] — *May 2024 – August 2024*
  Computing Scholar Intern — Livermore, CA, USA
  - Engineered an LLVM Attributor pass to **optimize byte layout within stack allocations** for improved spatial locality.
  - Implemented **LLVM IR** analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
  - Optimized the **LLVM GPU sanitizer** by hoisting redundant checks out of loops, reducing XSBench runtime by **6%**.
  - Built an LLVM IR basic-block permutation pass to evaluate performance sensitivity to code layout.
- **Nod.ai Labs** [] — *May 2022 – August 2022*
  Machine Learning Compiler Engineering Intern — Santa Clara, CA, USA
  - Contributed to **Torch-MLIR**, a compiler stack for lowering programs from the **PyTorch ecosystem to MLIR**.
  - Implemented kernels and compiler support for multiple **PyTorch operators** in the Torch-MLIR pipeline.
  - Added support for the **Facebook DLRM** recommendation model in the PyTorch-to-MLIR compilation pipeline.
  - Investigated improvements to the **ILP solver** used by the Alpa scheduling framework.
- **Astrome Technologies Private Limited** [] — *May 2019 – August 2019*
  Engineering Intern — Bengaluru, Karnataka, India
  - Integrated an eMMC IP block with an AXI-Lite-to-Wishbone bridge using a SystemVerilog wrapper.
  - Verified the eMMC IP and bridge integration through Vivado behavioral simulations.
  - Built XSLT and Python tooling to render XML data as internal HTML reports.
  - Implemented convolutional encoding and Viterbi decoding algorithms in C for error correction and detection.
  - Achieved an **order-of-magnitude speedup** over the original Python implementation.
- **Purdue University** [] — *May 2021 – Present*
  Graduate Research Assistant — West Lafayette, IN, USA
  - **Gibbon Compiler**
    * Optimized layouts of recursive datatypes to improve spatial locality.
    * Measured speedups of **1.14x to 54x** over the best prior work.
    * Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.
    * Structure-of-arrays transformation enables selective traversal, our benchmarks saw speedups of **1.1x to 12x**.

        * Developing optimizations enabled by the SoA representation such as loopification and vectorization.
    - **Orchard Compiler**
        * Developed a compiler framework to automatically fuse and parallelize tree traversals.
        * Measured speedups ranging from **1x to 5x** over baseline tree traversals.
    - **Copse Compiler**
        * Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.
    - **Cornucopia**
        * Developed a fuzzing-based framework to generate a corpus of optimized binaries from source programs using `llc` optimization flags.
        * Discovered bugs in LLVM compiler optimization passes.
        * Exposed decompilation failures in angr, Ghidra, and related binary analysis tools on generated binaries.
        * Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools (BATs).

- **Senior Design Project, SoCET Team [🏛]** *July 2020 – May 2021*
  Senior Design Team Member; advised by Dr. Mark Johnson and Prof. Samuel Midkiff West Lafayette, IN, USA
    - Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
    - Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
    - Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

Selected Projects

- **Global Tensor Layout Optimization for IREE** *June 2026*
  Researched on techniques for global tensor layout optimization in the IREE compiler pipeline.
- **LLM-Assisted Specification Inference for Verus** *Fall 2025*
  Explored LLM-assisted inference of Verus specifications using symbolic reasoning tools.
- **OpenMP GPU Offloading for Parallel STL Algorithms** *Fall 2024*
  Built a proof-of-concept evaluation of OpenMP GPU offloading for C++17 STL-style algorithms.

Publications

C=Conference, J=Journal, T=Thesis

[C.1] Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni *"Vectorized Secure Evaluation of Decision Forests"* in *PLDI 2021* [ACM DL]

[C.2] **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry *"Cornucopia: A Framework for Feedback Guided Generation of Binaries"* in *ASE 2022* [ACM DL]

[C.3] **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni *"Optimizing Layout of Recursive Datatypes with Marmoset"* in *ECOOP 2024* [LIPIcs]

[J.1] **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni *"Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals"* in *TACO 2024* [ACM DL]

[C.4] Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton *"Garbage Collection for Mostly Serialized Heaps"* in *ISMM 2024* [ACM DL]

Coursework

Compilers	Compiler Code Generation and Optimization; Compilers for GPUs; Introduction to Compilers and Translation Engineering
Programming Languages	Reasoning About Programs; Programming Languages; Advanced C Programming; Object-Oriented Programming in C++
Algorithms & Theory	Computation Models and Methods; Applied Algorithms; Data Structures in C; Formal Languages, Computability, and Parallelization
Systems	Computer Architecture; Programming Parallel Machines; Microprocessors and Microcontrollers; Digital Design
Mathematics	Discrete Mathematics; Graph Theory; Category Theory; Random Variables
AI & Data Science	Introduction to Artificial Intelligence; Python for Data Science
Security & Quantum	Holistic Software Security; Introduction to Quantum Computing

Skills

Programming Languages	C; C++; Haskell; Python; Java; Bash; MATLAB; Assembly (ARM, x86); SystemVerilog
Compiler & Verification	LLVM; GCC; Coq; Dafny; Pulse; GDB
Parallel & HPC	CUDA; OpenMP; MPI; Pthreads; OpenCilk; Intel TBB; Intel Cilk Tools
Data & Machine Learning	PyTorch; TensorFlow; NumPy
Systems & Databases	Unix/Linux; Windows; PostgreSQL; AFL++; gem5
Web & Development Tools	HTML; CSS; Git; Visual Studio Code; Vim; CLion; Notepad++; Kate
Professional Skills	Communication; teamwork; inclusive collaboration

Teaching Experience

- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Fall 2025*
- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Spring 2023*
- **Undergraduate Teaching Assistant**, Prof. Cheng Kok Koh, ECE 368, Purdue University *Spring 2020*
- **Student Grader for Signals and Systems**, Prof. Chih-Chun Wang, ECE 301, Purdue University *Spring 2020*

Talks Given

- **MWPLS 2025**, AoS To SoA Transformation of packed ADTs *Fall 2025*
- **MWPLS 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **ECOOP 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **PurPL Seminar**, Optimizing Layout of Recursive Datatypes with marmoset *Spring 2024*
- **ASE 2022**, Cornucopia: A Framework for Feedback Guided Generation of Binaries *Fall 2022*

Honors and Awards

- **Received a travel grant**, LLVM developers meeting *October 2024*
- **Semester Honors**, Purdue University *Fall 2019 - Spring 2020*
- **Deans's List and Semester Honors**, Purdue University *Fall 2017 - Spring 2018*

Leadership Experience

- **Seminar Co-Coordinator**, Purdue Programming Languages and Systems Research Group (PurPL) *Fall 2025 - present*

Review Experience

- **Member of the Artifact Evaluation Committee (AEC)**, ICFP 2026 *Summer 2026*
- **Member of the Program Committe (PC Member)**, C3PO 2026 *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, PLDI 2026 *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, SPLASH 2026 *Spring 2026*
- **Additional Reviewer**, PPoPP 2026 *Fall 2025*
- **Member of the Reviews for the Journal of Open Source Software**, JOSS *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, POPL 2026 *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, CGO 2026 *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, ICFP 2025 *Summer 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, PLDI 2025 *Spring 2025*
- **Collaborative Reviewer with Advisor (Milind Kulkarni)**, CGO 2025 *Spring 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, CGO 2025 *Fall 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, PPoPP 2024 *Spring 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, PPoPP 2023 *Spring 2023*

Volunteer Experience

- **Talk moderator**, LLVM developers meeting 2024 *Fall 2024*
- **Student Volunteer**, ECOOP/ISSTA 2024 *Spring 2024*
- **Student Volunteer**, SPLASH 2021 *Fall 2021*

Professional Memberships

- **Eta Kappa Nu**, Purdue Electrical and Computer Engineering Honors Society *Fall 2018 - Present*

Mentoring

- **Logan Anderson**, Phd @ Purdue *Fall 2025*
- **Aniruddh Srivastava**, Undergrad @ Purdue *Fall 2025*
- **Peter A Kaya Gretchikha**, Undergrad @ Purdue *Fall 2024 - Fall 2025*
- **Mikah Kainen**, Undergrad @ Purdue *Fall 2023*
- **Qingyuan (Peter) Li**, Undergrad @ Purdue, SURF summer research mentor *Spring 2023 - Fall 2023*